



Amir P. Adam

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ABOUT ME

ECE graduate with practical experience in embedded firmware, IoT system integration, and edge computing. Worked with ESP32, STM32, and Raspberry Pi on real projects involving C/C++, Python, MQTT, and REST APIs. Strong debugging instincts and a track record of delivering complete, working systems.

EDUCATION AND TRAINING

16/08/2022 – CURRENT Mohali , India

BACHELOR OF ENGINEERING IN ELECTRONICS & COMMUNICATIONS Chandigarh University

Relevant Coursework: Embedded Systems, VLSI Design, Communication Systems, Digital Signal Processing

Level in EQF EQF level 6

SKILLS

Programming

C/C++ | Python | Typescript

Embedded Systems

Microcontrollers: ESP32,STM32 | Raspberry Pi | Sensor Integration | Communication Protocols: I2C,SPI,UART

Tools &. Platforms

STM32CUBEIDE | PlatformIO | Arduino IDE | VSCode | Git | GitHub

LANGUAGE SKILLS

Mother tongue(s): **ARABIC(SUDANESE DIALECT)**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PROJECTS

02/04/2026 – 22/04/2026

IoT Food Monitoring System:Cloud Analytics & Mobile Dashboard

ESP32, DHT11, MQ135, AWS IoT Core, MQTT/TLS, AWS Lambda, Firebase Cloud Firestore & FCM, React Native | May 2026

- Architected secure edge-to-cloud pipeline: ESP32 transmits DHT11/MQ135 sensor data via **TLS-encrypted MQTT** to AWS IoT Core with **<<2 s** ingestion latency.
- Built serverless AWS Lambda backend to parse, validate, and compute food freshness status; wrote real-time/historical data to Cloud Firestore and triggered **FCM push alerts** to React Native app.
- Delivered cross-platform mobile dashboard with real-time Firestore sync, achieving **<<500 ms** UI update latency from sensor threshold breach to user notification.

Link <https://github.com/amirphiladam2/IoT-Based-Food-Monitoring-System>

05/03/2026 – 19/03/2026

Edge AI Surveillance System: Real-Time Intrusion Detection

Raspberry Pi 5, Python, OpenCV, TensorFlow Lite, FastAPI, Telegram Bot API | Mar 2026

- Achieved **~15 FPS** real-time inference on Raspberry Pi 5 using quantized MobileNet SSD, maintaining **>85% detection confidence** for human targets.
- Reduced alert latency to **<<2 s** via multithreaded asynchronous Telegram notifications with annotated snapshots and metadata.
- Served live MJPEG streams and logged **100+ detection events** in SQLite with timestamps via a FastAPI web interface.

Link <https://github.com/amirphiladam2/edge-ai-surveillance>

20/10/2025 – 20/11/2025

Disaster-Resilient Offline Mesh Communication System

ESP32, LoRa | Oct – Nov 2025

- Established decentralized peer-to-peer messaging across **>500 m line-of-sight range** using ESP32 + LoRa without cellular or internet infrastructure.
- Cut active power consumption by **~35%** via optimized duty-cycling and deep-sleep firmware, extending battery runtime in emergency scenarios.
- Achieved **<<3 s** hop-to-hop relay latency for situational awareness packets with integrated local sensor acquisition.

Link <https://techtacklenerd.blogspot.com/2025/11/disaster-resilient-hybrid-mesh.html>

● CREATIVE WORKS

12/07/2023 – CURRENT

YouTube Tech Content Creator

- Created technical educational content focused on Embedded Systems and App development concepts.
- Produced video tutorials demonstrating real-world implementations using STM32, ESP32, React Native.

Link <https://www.youtube.com/@amirdevstudios/videos>